

5. Describe the physiological effects of a high-flow nasal cannula device (HFNC) e.g. Optiflow™ or Airvo™. (30%). Briefly list the indications, and justify its use in an adult endoscopy suite (70%).

(Modified from ANZCA 2018A)

Examiner Report

Pass 60.6% Mean score 5.1

This question was answered reasonably well. There needed to be a basic understanding of physiology including **humidification** and the generation of **CPAP through high flow**. A reasonable argument for HFNC use (e.g. hands-free airway management, increased margin of safety in complex patients) needed to be made. Candidates who didn't do well often only mentioned apneic oxygenation and failed to discuss its use in preoxygenation or the spontaneously ventilating patient.

(a) Physiological effects of HFNC include

Respi

- Allows for High FiO₂
 - O₂ reservoir by washing out nitrogen in FRC and dead spaces
 - Apnoeic oxygenation which prolongs apnoeic time
 - May result in resorption atelectasis
- Passive ventilation by means of bulk flow/ Pendelluft effect
 - Improves carbon dioxide clearance
- Warming and humidification of airways
 - Improves secretion clearance and ciliary function
- CPAP effect from high air flows
 - Improved V/Q ratio from alveolar recruitment and decreased atelectasis.
 - Airway splinting effect maintains airway patency, and reduces work of breathing
 - Risk of barotrauma if excessively high airway pressures or in emphysematous lungs

CVS

- Afterload may improve from reduced pulmonary congestion/ oedema from CPAP
- Venous return/ preload may worsen from positive intra-thoracic pressures generated from CPAP
- Effect on cardiac output is variable, depending on the interplay between afterload vs preload effects

GI

- Aerophagia from excessive positive airway pressure, especially in patients with GERD/ reduced lower oesophageal sphincter tone.
- Risk of aspiration secondary to regurgitation from aerophagia

(b) Indications for HFNC

- Type 1 Respiratory Failure
 - Cardiogenic pulmonary oedema
 - Non-cardiogenic T1RF
- Pre-oxygenation/ apnoeic oxygenation for
 - High risk group (morbidly obese patient)
 - Awake fibre-optic intubation
 - Bronchoscopic procedures

- Transitioning from invasive ventilation in ICU
- Post-extubation NIV to reduce atelectasis/ post-op hypoxemia for high risk patients (cardiac, major abdominal surgery, OSA)

(c) Justify its use in adult endoscopy suite

HFNC provides high FiO₂, passive ventilation, CPAP, and warm and humidified gases in a compact, portable package, which is useful in a remote location with confined spaces such as an endoscopy suite.

For procedures done under sedation/monitored anaesthetic care, HFNC is useful because it:

- Allows for delivery of higher FiO₂ without interfering with procedure (compared to non-rebreather mask NRM)
- Maintains airway patency during sedation/ MAC
- Allows for hands-free airway management
- Increases margin of safety for desaturation/ respiratory compromise in complex patient group
- Improves patient comfort and tolerance via delivery of warm, humidified gases

For procedures done under GA with intubation, HFNC is useful because it:

- Facilitates pre-oxygenation, and prolongs apnoeic time prior to intubation. This is important in a remote location where help may be difficult to get.
- Allows for post-extubation NIV in complex patient group, and reduces re-intubation rate in recovery room post-procedure

Caution with HFNC use in the endoscopy room for

- Patients at high risk of aspiration/reflux, which could be the indication for scopes.
- Patients with severe cardio-respiratory disease. It would be safer to have these patients done in major OT with appropriate facility and staff support
- If staff are inexperienced or unfamiliar with its use
- May lead to false sense of security as SpO₂ may appear to be satisfactory even though patient is deteriorating